Construction and Building Materials 315 (2022) 125738

Contents lists available at ScienceDirect

Construction and Building Materials

journal homepage: www.elsevier.com/locate/conbuildmat

Life cycle assessment of asphalt and cement pavements: Comparative cases in Shanxi Province

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A R T I C L E I N F O

Keywords:
Life cycle assessment
Energy consumption
Environmental pollutant emission
Freshwater toxicity

A B S T R A C T

In order to understand the energy consumption and environmental pollution level of Shanxi highway construction, the life cycle environmental and energy impact assessment of asphalt mixture and cement concrete were conducted. One representative cement concrete pavement and two asphalt mixtures highways were selected for quantitative evaluation and LCA analysis. The total energy consumption values of Taichang highway are 20 % higher than that of the Dayun highway. The global warming potential indicators of Taichang (asphalt mixture), Dayun (asphalt mixture), and Deda (cement concrete) are 181 kg eq.CO_2, 285 kg eq.CO_2, and 330 kg eq.CO_2, respectively during the materials production stage. The results indicated that the environmental effects of global warming had the biggest impact on the life cycle of the asphalt mixtures and the respective highways. Based on the study results, environmental pollution was highest on the Taichang highway whilst the Deda highway had the highest values of freshwater toxicity. Furthermore, energy consumption during the construction stage and environmental impacts at the end-of-life stage were found to be alarmingly high. Thus, reducing the energy consumption during the material preparation and highway construction stages as well as controlling the environmental pollutants and emissions during service are key to conserving energy, preserving the surrounding environment, and optimizing the life cycle of Shanxi highways.

1. Introduction

With the economic development, the transportation development of Shanxi Province has met a tremendous development opportunity. As of 2020, Shanxi Province had a total of 30 major highways. Taichang (Taiyuan to Changzhi and Jincheng), Dayun (Taiyuan to Datong and Yuncheng) and Deda (Deshengkou to Datong city) are the three principal highways of Shanxi Province. Due to the high energy consumption and construction/maintenance costs associated with these highways, it is necessary to comparatively evaluate and quantify their sustainability (both economically and environmentally) through conducting a life cycle assessment (LCA) [1–2].

LCA can be used for evaluating potential environmental pollutants and energy consumption throughout a product or structure's life [3–4]. Recent studies have shown that energy consumption at the usage stage of asphalt pavements is affected by various factors including materials type, functional layer structure and pavement thickness [5–6]. Aside from the associated costs and high potential for resources depletion, energy consumption will also lead to severe greenhouse (CO_2) emissions. Reducing CO_2 emissions in the transportation field has thus become one of the key remedies of mitigating national carbon emissions for many countries [7].

Numerous studies have been carried out in China to evaluate the environmental impacts of asphalt pavements using LCA in China [8]. From their findings rubber modified asphalt pavements showed high environmental impacts compared to reclaimed asphalt pavements [9]. Similarly, AzariJafari et al. [10] studied the human toxicity potential, metal release and the hazardous waste of recycled asphalt pavements and found that the contaminant (Cd, Cr, Se and Ag) transported from the bottom ash resulted in a very low unsaturated zone. The "cradle-to-

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https://doi.org/10.1016/j.conbuildmat.2021.125738
Received 14 June 2021; Received in revised form 10 November 2021; Accepted 16 November 2021
Available online 21 November 2021

grave" LCA always involves many stages and activities that result in different energy and environmental paradigms [11–12]. Asphalt pavement preservation such as use of thin overlays is not only cost-effective, but also brings significant environmental benefits [13–14]. Likewise, use of bio-asphalt materials could significantly reduce the energy consumption factor and environmental pollutants [15]. The literature shows that CO_2 emissions and environmental impacts related to the roads and highways are just one magnitude lower than that of bridges and tunnels [16]. Asphalt pavement can affect significantly the environment and economy of China [17–19]. Although there are many researches about LCA of asphalt pavement, there are lack of LCA of cement concrete pavement, and LCA analysis method of Shanxi highways based on the various stages of energy and environmental assessment are not clear.

Based on the above background, as a step towards minimizing energy consumption and reducing the environmental pollution, the LCA of representative highways in Shanxi Province, were comparatively evaluated in this study, with a focus on quantifying the energy and environmental impacts. The functional units, boundary conditions, life cycle inventory and data acquisition of LCA of Shanxi highway were proposed. LCA energy consumption evaluation included impacts of different highways and pavement types and impacts of different highway life cycle stages were investigated. Further, a life-cycle analysis method was also formulated based on the various stages of energy and environmental assessment. This study proposed the four environmental impact indicators (Global warming potential, Acidification potential, Freshwater toxicity, and Human toxicity) and its detailed evaluation compounds, which were used to verify the environmental impact indicators whether were suitable with the typical highways in Shanxi Province.

2. Study methodology

The LCA in this study was conducted in accordance with the ISO standard 14040/14044 [20] using the 2019 Chinese energy data. Three major highways, namely Taichang, Deda, and Dayun, respectively, in Shanxi Province of China were selected and used as the case study. The respective highway construction data were obtained from Shanxi Transportation Holdings Group CO., LTD.

As discussed in the subsequent text, the study methodology incorporated defining the scope of work and the corresponding research tasks. The study functional units, boundary conditions, and highway pavement structures are also discussed in this section of the paper.

2.1. Scope of work and research tasks

LCA is a useful tool for evaluating the energy consumption and the impacts of environmental pollution. Within this framework, the scope of work and research tasks for this study involved addressing the following objectives:

- Energy consumption assessment, mainly focusing gasoline and diesel usage;
- Evaluation of the environmental impacts of each stage of the LCA which included CO_2, CO, CH_4, SO_2, NO_x, NMVOC and PM2.5 emissions; and
- Identification of the potential environmental improvements by analyzing different types of asphalt mixture pavements and cement concrete pavements from a life cycle point of view.

As previously mentioned, the selection of the highway types to be studied was based on considering that there are three major highways in Shanxi Province (China), namely:

a) Taichang (asphalt mixtures pavement),
b) Deda (cement concrete pavement) and
c) Dayun (asphalt mixtures pavement)

2.2. Study functional units and boundary conditions

The selected study functional units, which represents the fundamental unit throughout the whole LCA study, were each 1.0 km length of a highway pavement section. The pavement layer thicknesses for the different highways were different and were evaluated as such. The study included all the stages and activities that encompass materials production, transportation, construction, operations and maintenance work, and end-of-life during the life cycle process as per the "cradle-to-grave" method [21]. The "cradle-to-grace" method is that everything ranges from raw material acquisition to production, use and disposal.

The total pavement width for each of the three selected highways (namely Taichang, Deda, and Dayun), incorporating all the lanes, was about 27.5 m. The corresponding pavement structures, mostly comprising of the surface course, binder course, unbound base course, and subgrade course, are shown in Table 1.

Other highway structures such as embankments, drainage systems and road markings were not included in this study because their contributions is negligible for highway construction and LCA [22].

The pavement surface layer thickness of the Taichang, Deda, and Dayun highways are 17 cm, 28 cm, and 16 cm, respectively. The boundary conditions and the flow sequence adopted for the study are shown in Fig. 1.

From Fig. 1, the flow sequence for LCA and the boundary conditions can be summarized as bullet-listed below:

- Materials production including raw materials and energy supplements.
- Transportation including raw materials transportation, earthwork removal/transportation and equipment transportation.
- Construction, which is made up of the pavement and base, or sub-base construction and includes all the stages of the road construction.
- Operations and maintenance work including maintenance, demolition, and reconstruction of the pavement, in order to ensure functionality, in terms of the bearing capacity, surface layer or thin overlay over the lifespan of the pavement construction.
- End-of-life, which consists of recycling, reuse, or disposal of solid waste, or regenerated material.

To sustain and maintain the average life span of road pavements, there is often a need (including preventive maintenance) to ensure suitable levels of service. The fundamental data of road pavement showed as Table 2 and these data obtained from Chinese Reference Life Cycle Database (CLCD) [23]. The database is that a national background LCI database consists of ca. 600 LCI datasets for key materials and chemicals, energy carriers, transport, and waste management. In this study, the useful lifetime of Taichang and Dayun highway pavements (asphalt) was assumed to be 15 years while that of Deda highway

Table 1
Highway pavement structures.

Structure layer	Taichang	Deda	Dayun
	Material types		
Upper layer	4 cm SMA-13 (asphalt)	28 cm cement concrete	4 cm SMA-13 (asphalt)
Middle layer	6 cm AC-20	–	5 cm AC-16
Lower layer	7 cm AC-25	20 cm cement stabilized macadam	7 cm AC-20
Base	33 cm cement stabilized macadam	36 cm ash stabilized soil	20 cm cement stabilized macadam
Sub-base	20 cm composite stabilized soil	20 cm gravel	20 cm cement stabilized macadam

Legend: AC = asphalt concrete, SMA = stone matrix (mastic) asphalt.

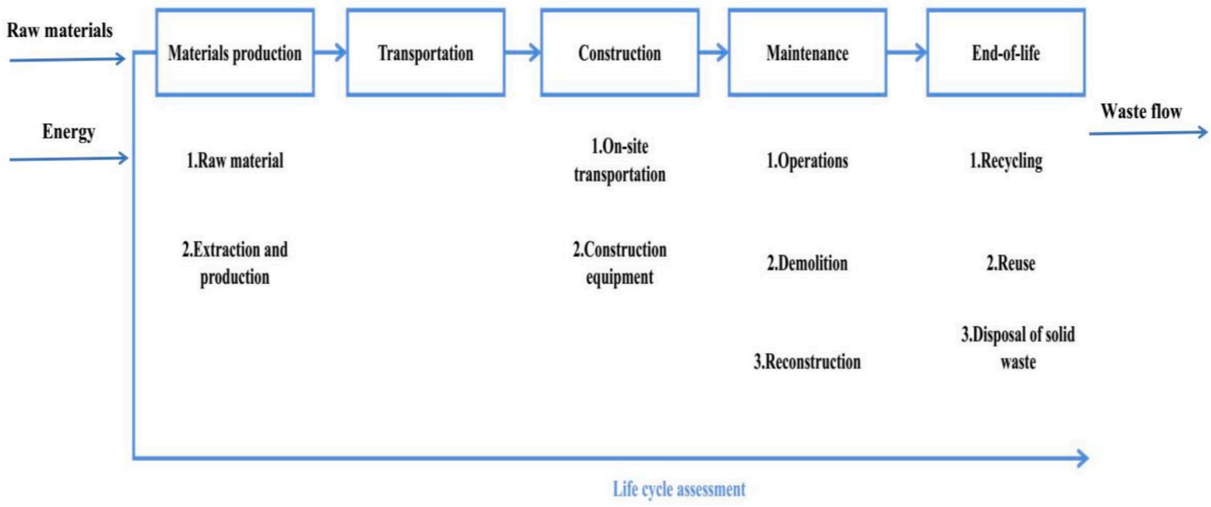

Fig. 1. Life cycle stages diagram.

Table 2
The fundamental data of road pavement.

	Data of raw materials	
Raw materials	Energy consumption (MJ/t)	Environmental pollutants (kg/t)
Base asphalt	2830.760	189.120
SBS modified asphalt	5412.220	232.040
Rubber	3.590	0.970
Aggregate	31.990	$2.380*10^{-3}$
Cement	3400.000	79.850
Data of materials production (mixtures production)		
Mixture type	Energy consumption (MJ/t)	Environmental pollutants (kg/t)
SMA-13	336.670	28.260
AC-13	334.990	27.810
AC-16	325.860	28.050
AC-20	334.990	27.810
AC-25	307.800	28.260
Data of Transportation stage		
Transportation methods	Energy consumption (MJ/t)	Environmental pollutants (kg/t)
30 t heavy diesel truck	0.804	0.075
Data of construction stage (Paving process of mixtures)		
Mixture type	Energy consumption (MJ/t)	Environmental pollutants (kg/t)
SMA-13	15.860	1.180
AC-13	13.220	0.980
AC-16	11.580	0.750
AC-20	8.810	0.660
AC-25	7.340	0.550
Data of construction stage (Rolling process of mixtures)		
Mixture type	Energy consumption (MJ/t)	Environmental pollutants (kg/t)
SMA-13	18.600	1.380
AC-13	13.230	0.980
AC-16	12.850	0.960
AC-20	12.480	0.930
AC-25	10.390	0.770
Data of maintenance stage		
Mixture type	Energy consumption (MJ/t)	Environmental pollutants (kg/t)
SMA-13	18.290	306.350
AC-13	15.970	267.520
AC-16	15.250	255.850
AC-20	14.800	247.890
AC-25	7.970	119.30

pavement (concrete) was assumed to be 20 years. Maintenance stage, maintenance or preventive maintenance of the top layer of the pavement structure included an ultra-thin wearing course or skid-resistant surface layer, which was assumed to occur at the third of the asphalt pavements' service life (i.e., per 5 years/maintenance).

3. LCA formulation and data assembly

The LCA formulation including the software method used and the environmental impact indicators considered in the study are discussed in this section of the paper. The life cycle inventory and data acquisition for LCA are also discussed in this section.

3.1. LCA method used and environmental impact indicators

Life cycle assessments were computed using the materials balance method whilst the life cycle environmental impacts indicators were formulated using the ReCipe method and OpenLCA LCIA methods 1.5.7. In this study, the LCA was evaluated using the openLCA1.10.3 software [24]. Life cycle energy consumption and emission data were used based on the database of CML 2015. The fundamental energy consumption lists included carbon emission values (g/MJ), specific energy consumption (Q_{net}, MJ/kg), and emission factors (g/t). The eco-indicator for climate change was CO_2 emission while the eco-indicator for air acidification was SO_2.

The main eco-indicator of human toxicity were NO_x, CO, NMVOC, PM2.5, NH_3 and heavy metals which can be represented by 1,4-dichlorobenzene equivalent. The eco-indicator for freshwater toxicity was hydrocarbon and heavy metal which could be represented by 1,4-dichlorobenzene equivalent. The eco-indicator for terrestrial eco-toxicity was part of NMVOC, hydrocarbon and heavy metals, which could be represented by 1,4-dichlorobenzene equivalent. In this study, the energy and environmental indicators or categories were evaluated and showed as follows:

- Global energy consumption (MJ);
- Global warming potential (GWP, kg eq CO_2);
- Acidification potential (AP, kg eq SO_2);
- Freshwater toxicity (FT, kg eq 1,4-dichlorobenzene included hydrocarbon and heavy metals); and
- Human toxicity(HT, kg eq 1,4-dichlorobenzene included NMVOC + PM2.5 + NO_x)

3.2. Life cycle inventory and data acquisition

Life cycle inventory (LCI) was conducted to quantitatively evaluate the energy and environmental inputs and outputs of the LCA system. The

LCI data is shown in Table 3.

The specific energy consumption of natural gas, crude oil, coal, gasoline, and uranium are 48.0 MJ/kg, 42.3 MJ/kg, 26.7 MJ/kg, 45.3 MJ/kg, and 8.5*10^{-4} MJ/kg, respectively. The specific gravity of the raw material for (1 t) asphalt production were 20.1 kg/t (natural gas), 40.9 kg/t (crude oil), 1.03 kg/t (coal), 25.8 kg/t (gasoline), and 0.0001 kg/t (uranium), respectively. The energy consumption was calculated as the cumulative energy demand (MJ). The inventory data for the raw materials were obtained from the Ecoinvent database (Version 3.6).

Asphalt mixture or cement concrete pavement process is not only energy energy-consumption, but also includes environmental pollutants emissions such as CO_2, CO, CH_4, SO_2, NO_x, NMVOC and PM2.5. The LCI of environmental pollutants are showed in Table 4.

The emission factors of the environmental pollutants listed in Table 4 are very important for evaluating the LCA of the pavement. The total energy consumption of the raw materials for all the stages of the LCA are shown in Table 5.

The different highways had different data of raw material consumption. These data were the basis for calculating the total energy consumption and environmental pollutants. The fundamental data for energy requirements of the machinery during mixture production, construction (mixing), and transposition were all obtained from the standard of JTGT 3832–2018 “Highway Engineering Budget Quota”.

The fuel consumption quota of the construction machinery of 1 work day is shown in Table 6. Using this table, the total fuel consumption and energy consumption of the construction machinery were calculated. Thereafter, the environmental pollutants arising from construction machinery were estimated.

4. Results and discussions

The LCA results are presented, analyzed, synthesized, and discussed in this section of the paper. These results and analyses include energy consumption and environmental impact evaluations.

4.1. LCA energy consumption evaluation

The energy consumption evaluation included LCA for different highways and life cycle stages. These results for these LCA evaluations with respect to energy consumption are presented and discussed in this subsection of the paper.

4.1.1. Impacts of different highways and pavement types

The life cycle energy consumption impacts of different pavement types and different stages on emission were analyzed and compared to the traditional pavements. Reducing energy consumption and decreasing environmental pollution constitute some of the key measures to ensure sustainable development [25]. The energy consumption of the highways is shown in Fig. 2.

As shown in Fig. 2(a), the total energy consumption of the Deda highway is smaller than that of Taichang highway and Dayun highways. These results demonstrate that Deda highway, with a cement concrete pavement structure had the smallest energy consumption and that cement concrete can be rated as a low energy consumption pavement technology when compared to asphalt.

Furthermore, the total energy consumption values of Taichang highway is greater than that of Dayun highway. According to the highways construction experts, Taichang highway and Dayun highway are comprised of almost similar pavement structures, particularly the upper surfacing layer - see Table 1. Therefore, they should ideally have similar total energy consumption. However, the results (Fig. 2(a)) show that the total energy consumption values of Taichang highway are 20 % higher than that of the Dayun highway. As evident in Table 1, the main reason for this is that the middle and base layer thickness of the Taichang highway are greater than that of Dayun highway, leading to an increase in then energy consumption.

According to the literature, the total average energy consumption of Chinese highways is 152524.42 MJ [26]. Compared to this study, the highways in Shanxi Province consume a lot of energy. The main reason is the unprofessional supervision [27]. To mitigate this situation, the government can use a public tender and impose energy consumption policies/regulations on designers, builders, inspectors, and clients. Unified training for the construction team can also be mandated.

4.1.2. Impacts of different highway life cycle stages

The life cycle was divided into five stages that included materials production, transportation, construction, operations/maintenance work, and end-of-life. As shown in Fig. 2(b), the energy consumption for materials production is the largest whilst the operation and maintenance stage has the smallest energy consumption values. Due to low energy consumption values, the numerical column for the operation and maintenance stage in Fig. 2(b) almost invisible.

From Fig. 2(b), it can also be observed that the combined energy consumption ratio for materials production and transportation stages

Table 3
LCI data for raw materials and energy.

Energy	Q_{net}(MJ/kg)	Asphalt (kg/t)	Cement (kg/t)
Natural gas	48.0	20.1	–
Crude oil	42.3	40.9	–
Coal	26.7	1.03	96.0
Gasoline	45.3	25.8	–
Uranium	8.5*10^{-4}	0.0001	0.0005
Eurobitume	3227*10^{-3}	–	–
Total energy consumption of asphalt production	4900*10^{-3}	–	–

Table 4
LCI of emissions factors for the environmental pollutants

Emission factor	Asphalt (g/t)	Cement (g/t)
CO_2	174,244	605,000
SO_2	781	113
NO_x	770	165
NMVOC	331	305
PM2.5	161	95

Table 5
Fuel consumption quota of the construction machinery for 1 work day

Type	Taichang	Dayun	Deda
Crude oil (kg)	12686.08	11452.15	125.58
Gasoline	108.65	106.26	215.25
Diesel oil	576.61	498.78	1251.68
Electricity (kW·h)	8089.02	7856.46	9125.65

Table 6
Total energy consumption of the raw materials and machine values

Type	Taichang	Deda	Dayun
Work day (d)	162.30	252.12	215.25
Fine aggregate asphalt mixture (t)	3060.00	0	3215.00
Cement concrete (t)	0	4250.68	0
Asphalt (t)	343.60	0	361.51
Fillers (t)	297.24	0	308.68
Aggregate chips (t)	1169.32	1862.15	1202.23
Crushed aggregate (t)	3361.52	6215.48	3625.15
Wheeled loader (d)	19.77	25.15	21.02
Mixing equipment (d)	7.40	6.25	7.80
Paving equipment (d)	8.79	5.48	9.15
Road roller (d)	50.07	12.46	51.02
Dump truck (d)	7.99	6.22	8.12
Sprayer truck (d)	1.23	12.84	1.25

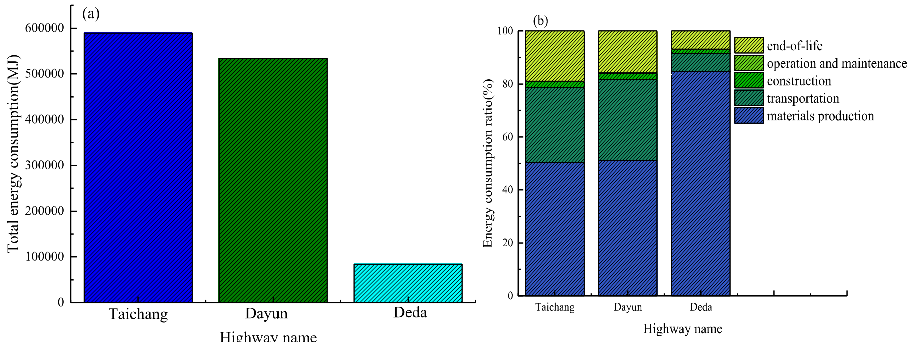

Fig. 2. Energy consumption: (a) highway total energy consumption, (b) energy consumption at different life cycle stages.

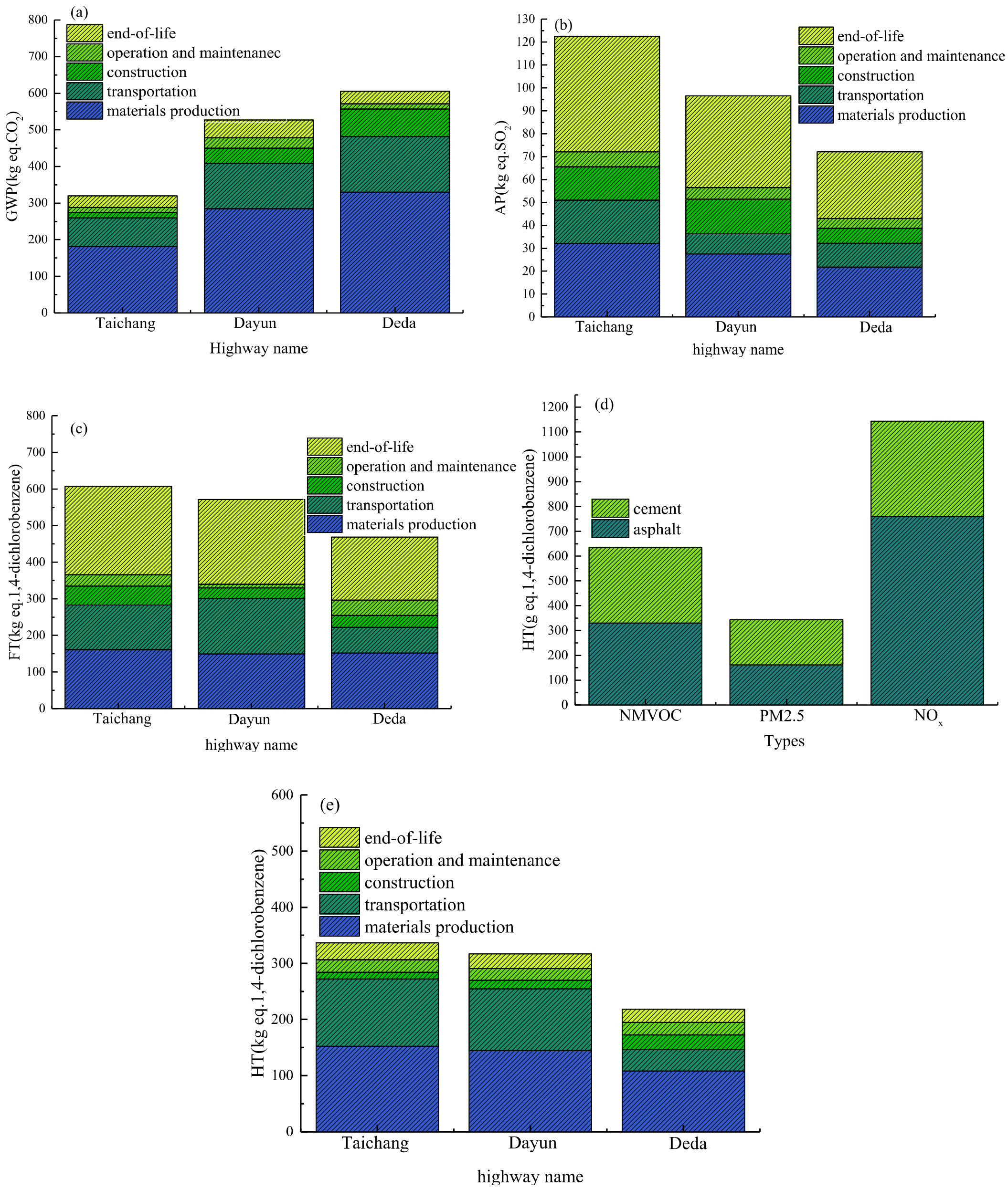

Fig. 3. Environmental pollution: (a) highway global warming potential (GWP = Global Warming Potential), (b) highway acidification results (AP = Acidification Potential), (c) Highway freshwater toxicity (FT = Freshwater Toxicity), (d) human toxicity (HT) of materials, (e) human toxicity (HT) of highway comparisons.

was above 70 % of the whole life cycle of the highways. This illustrates that transportation and material production stages are the primary energy consumers during the life cycle of these three Shanxi highways. Compared with energy consumption of Inner Mongolia highways [28], the primary energy consumers stages are different during the life cycle. However, the energy consumption ratios for the different stages were different in the three highways. Compared to the energy consumption of the Deda highway with respect to materials production, the energy consumption values of other two highways (i.e., Dayun highway and Changda highway) were small. However, the energy consumption with respect to the construction stage seem to be similar for all the highways.

Overall, the results in Fig. 2(b) indicate that the energy consumption of cement concrete-based highways is significantly higher than that of asphalt mixtures-based highways in terms of materials production. Furthermore, it can also be concluded that materials production is the main energy consumption stage in the cement concrete-based highways, such as Deda, followed by transportation and end-of-life stages.

Likewise, materials production was similarly found to be the largest energy consumer in the asphalt-based highways (but numerically lower than the concrete-based highways) consecutively followed by the transportation and end-of-life stages. For all the three highways, Fig. 2 (b) shows similar energy consumption for construction and negligible (i. e., less than 1.00 %) for the end-of-life stage. In comparison to asphalt-based highways, materials production and end-of-life stages were associated with the highest and least energy consumption, respectively, for concrete-based highways.

4.2. LCA environmental impacts evaluation

As discussed in the subsequent text, the environmental impacts evaluation included LCA for global warming, acidification potential, freshwater toxicity, and human toxicity.

4.2.1. Global warming potential (GWP) impacts

The factors responsible for global warming include truck, raw materials production, and paver emissions (CO_2) during the highway construction process. Minimizing CO_2 emissions is thus one of the key remedies to reduce global warming potential. The global warming potential (GWP) results are shown in Fig. 3(a).

As shown in Fig. 3(a), the global warming potential indicators of Taichang, Dayun, and Deda are 181 kg eq.CO_2, 285 kg eq.CO_2, and 330 kg eq.CO_2, respectively during the materials production stage. It is apparent from this figure that material production is the biggest potential CO_2 emission stage in as far as global warming is concerned. In terms of highway comparison, the global warming potential of Dayun highway was found to be the highest, whilst despite both being asphalt-based, Dayun exhibited greater global warming potential than Taichang highway. The probable reason for this is that the CO_2 emissions, as illustrated in Fig. 3(a), mainly arises from the raw materials production process, which, in the case of Deda highway has more carbon elements in the combined cement and asphaltic materials. By contrast, Taichang and Dayun highways are only associated with asphaltic raw materials.

Based on the materials balance perspective, there are various CO_2 emissions at the materials production stage. However, the three highways have a similar level and proportion of global warming potential at the each stage of the life cycle. The reason for this is that the total carbon source is the same for all the three highways.

Overall, it is apparent from Fig. 3(a) that concrete-based highways, such as Deda, are detrimentally associated with more global warming potential than asphalt-based highways. As theoretically expected, the materials production stage exhibited the greatest threat to global warming potential followed by the transportation stage. For all the three highways evaluated, operation and maintenance were the least detrimental stages to global warming seconded by construction and end-of-life stages for the asphalt-based and concrete-based highways, respectively.

4.2.2. Acidification potential (AP) impacts

As shown in Fig. 3(b), based on the life cycle impact assessment results, a hypothesis can be inferred that Dayun highway, at 50 % regenerated asphalt pavement and 3 cm of the wearing course, is the most environmentally friendly alternative among the three highways evaluated, with the lowest acidification potential (AP) values.

From Fig. 3(b), the total acidification potential values of Taichang highway, Deda highways and Dayun highway are 122.56 kg eq.SO_2, 72.18 kg eq.SO_2, and 96.52 kg eq.SO_2, respectively. These results indicate that Taichang highway has the highest quantitative value of acidification potential indicator and is not an environmentally friendly highway. Based on the relatively low values of Deda highway, Fig. 3(b) also indicates that cement concrete pavements has more potential to reduce acidification than asphalt-based highway pavements.

Compared to the acidification values of the different life cycle stages, the AP values associated with the end-of-life stage are the largest in magnitude, followed by the materials production stage. This means that the end-of-life stage is detrimentally more impactful in terms of highway acidification and is seconded by the materials production stage. On the other hand, the operation and maintenance stage exhibited the least highway acidification potential with the smallest AP values.

4.2.3. Freshwater toxicity (FT) impacts

Water is an essential natural occurring resource, particularly in freshwater [29]. Regrettably, freshwater toxicity (FT) is on an increasing trajectory and is continuously impacted by the activities of modern civilization including environmental pollutants arising from highways. Under sunlight irradiation, FT converts to an unstable state that is undesirably detrimental to the human health [30]. As illustrated in Fig. 3 (c), Taichang highway has the largest overall freshwater toxicity (FT) values closely followed by Dayun highway, with the end-of-life stage being the most impactful stage for all the highways during their life cycle. This is probably because the highway was built using asphalt mixtures and asphalt mixtures contain heavy metals and lots of hydrocarbons that potentially contribute to polluting the water – and hence, high overall FT values. This also may partly explain why the overall FT values of Taichang and Dayun highways, both asphalt-based pavements, are close to each and not significantly different.

From Fig. 3(c), it can also be observed that end-of-life is the most impactful stage with the highest FT proportional contribution for all the three highways. This is because high-content FT materials are usually wasted, demolished, and/or recycled at the end-of-life stage – thus, heavily contributing to freshwater pollution. On the other hand, Taichang highway and Dayun highways have almost similar overall FT values but different values for different stages of the life cycle. Although they are both comprised of asphalt-based highways, the slight differences in the pavement structure could be a contributing factor.

Whilst the total FT is relatively lower, the individual FT values for materials production stages and end-of-life stages of Deda highway (Fig. 3(c)) are comparable to those of asphalt-based highways, namely Taichang and Dayun. However, a significant difference exits for the construction stage with the asphalt-based highways being higher than the concrete-based Deda highway. Fig. 3(c) also shows relatively higher FT values for the construction, operation/maintenance, and end-of-life stages for the Taichang highway than Dayun highway and vice versa for the transportation stage. By contrast, the FT impact of materials products appeared to be similar is all the three highways evaluated. Overall, these FT results indicate that the materials production and end-of-life stages were the dominant sources of freshwater pollution and toxicity, followed by the transportation stage.

4.2.4. Human toxicity (HT) impacts

In this study, the human toxicity (HT) impacts were evaluated using the 1,4-dichlorobenzene equivalent method. 1,4-dichlorobenzene equivalent method used the NOx, NMVOC, and PM2.5 contents to evaluate the HT impacts. The bigger the NOx, NMVOC, and PM2.5

contents, the bigger the HT impacts. As shown in Fig. 3(d), the content of NO_x is the highest in magnitude and therefore, most impactful with respect to HT. By contrast, the contents of NMVOC and PM2.5 are comparatively small and therefore, hardly impactful with respect to HT. As shown in Fig. 3(e), the different highways have the different characteristics of HT impacts. Compared to Deda, Dayun and Taichang highways have the comparatively higher HT impacts. The reason for this may be that there are the emissions of VOC and NO_x in the asphalt mixtures-based highways. This is because there is a production of VOC and NO_x when the asphalt is heated above 100 °C. Overall, these results (Fig. 3(d-e)) indicates that compared to cement concrete-based highways, asphalt-based highways have comparatively greater HT impacts. The results (i.e., Fig. 3(d)) further demonstrate that asphalt is more detrimental to the human healthy than cement.

To reduce HT impacts, a recommendation can be made that the government gives more consideration to building the eco-friendly cement concrete-based highways. As shown in Fig. 3(e), the different stages of HT impacts in life cycle shows that the materials production and transportation stages are more impactful and severe than the other stages. Their combined impact proportion is over 85 %. There is a need to be cautious with the materials production and transportation stages during the highway construction process, especially for asphalt-based highways.

Fig. 3(d-e) indicates that there are some HT impacts in the operation and maintenance stage. These emissions mainly happens at the maintenance stage because this stage is associated with regenerating and heating the asphalt. The maintenance stage emits HT gas such as NMVOC and NO_x. The HT at the end-of-life stage is largely associated with the demolishing activities, pavement material removal, truck exhaust emissions, dismantling equipment, etc.

5. Conclusions

In this study, the life cycle assessment (LCA) of three representatives in Shanxi Province, namely Taichang, Dayun, and Deda, was comparatively evaluated. Based on the study results and findings, the following conclusions and recommendations were made:

- Deda highway had the smallest energy consumption indices, with its cement concrete pavement technology rating as a kind of low-energy consumption' pavement technology. The transportation stage and materials production and transportation stages were found to be the primary energy consumers during the life cycle of the highway. The combined energy consumption ratio for these two life cycle stages was over 70 %.
- The global warming potential indicator (Taichang = 181 kg eq.CO_2, Dayun = 285 kg eq.CO_2 and Deda = 330 kg eq.CO_2) had the largest CO_2 emission during the materials production stage. On the other hand, Taichang highway exhibited the greatest potential (AP = 122.56 kg eq.SO_2) for the acidification which suggested that Taichang is not an environmentally friendly highway. Further, the study findings indicated that cement concrete pavement had potential to reduce the acidification potential impact.
- Freshwater toxicity during the materials production stage and end-of-life stages were dominated on Deda highway and Taichang highways, whilst different levels of freshwater toxicity existed on Dayun highway. Compared to the cement concrete-based highway, asphalt based highways exhibited comparatively higher human toxicity impacts. The materials production and construction stages were found to play a significant role in the human toxicity impacts during the life cycle of highways, with combined HT ratio exceeding 85 %.
- Search for replaced or regenerate materials of petroleum-based asphalt;
- Generalize the new energy-saving paving technology.

CRediT authorship contribution statement

Xinxing Zhou: Writing – original draft, Investigation, Formal analysis. **Xiaorui Zhang:** Writing – review & editing. **Yuan Zhang:** Writing – review & editing. **Sanjeev Adhikari:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (No.52008235), and Special Foundation of Achievements Transformation Guide of Department of Science and Technology of Shanxi Province (No.201804D131034), and Science and Technology Project of Shanxi Transportation Holdings Group CO., LTD (No.19-JKKJ-55 and 20-JKKJ-43).

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